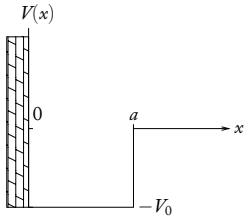
Problem Set 3

Due Saturday, September 26, 11:59 PM.

You may use Mathematica or other computational tools in all problem sets.

1. Consider a particle of mass m in one dimension in the potential shown in the figure below with $V(x) = \infty$ for $x < 0$, $V(x) = -V_0$ for $0 \le x \le a$ and $V(x) = 0$ for $x > a$.



   (a) Determine the transcendental equation for the energy of bound states in the well. (5 points)

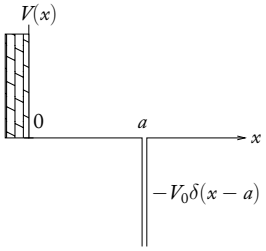
   (b) Determine the minimum value of V_0 which supports at least one bound state. (5 points)

   (c) Make a schematic drawing of the ground bound state wavefunction over the given potential. (5 points)

   (d) Determine the wavefunction for a particle with energy $E > 0$ in each region $0 \le x \le a$ and $x > a$. (5 points)

   (e) Determine the phase difference between the wavefunction in the region $x > a$ and the wavefunction in $0 \le x \le a$ for the particle in part (1d). What is the source for the phase difference? (5 points)

2. (a) Consider the potential shown in the figure below where $V(x) = -V_0\delta(x - a)$ for $x > 0$ and a wall with $V(x) = \infty$ for $x < 0$. This potential could be used as an approximation for an atom near a wall.



   i. Find a transcendental equation for the energy of a bound state. Determine the condition for which the particle is very far from the wall at $x = 0$ that the wall could be ignored.

(5 points)

ii. Determine the minimum value of V_0 that supports a bound state, in terms of other given parameters. (5 points)

3. How long did it take you to complete this problem set?